

Registering an Azure Resource Provider



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1. Purpose

Provide instructions for registering an Azure resource provider in a subscription so that the services and resource types offered by that provider can be deployed and managed.

Registration state affects deployments, which permissions are required, and how registration is performed step by step in the Azure portal.

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3. Registering a Resource Provider in an Azure Subscription

3.1. Background

Azure groups its services into provider namespaces such as Microsoft.Compute, Microsoft.Network, Microsoft.Storage, and Microsoft.Quota, and each namespace owns a set of resource types. When a resource is created interactively through the Azure portal, the portal will usually register the required provider automatically. Deployments driven by ARM templates, Bicep, Terraform, the CLI, or a service principal receive no such convenience, which is why a provider that has never been used in the subscription must be registered explicitly. Registration applies to the whole subscription rather than to a resource group, is not inherited from a management group, and must be repeated in every subscription that needs the provider.

3.2. Prerequisites

The following are required before a provider can be registered:

- Access to the subscription that will host the workload, and the subscription ID or name.
- An account with the **Owner** or **Contributor** role at the scope of the subscription.

3.3. Registering a Provider in the Azure Portal

The Azure portal is the standard method for registering a provider. It requires no tooling on the workstation, shows the registration state of every provider available to the subscription, and gives

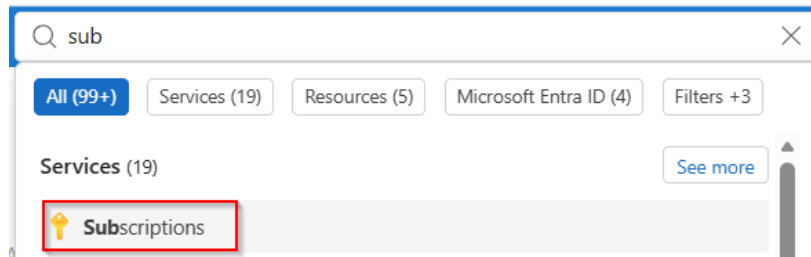
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immediate visual confirmation that the change took effect. The steps below register Microsoft.Quota; the same sequence applies to any namespace.

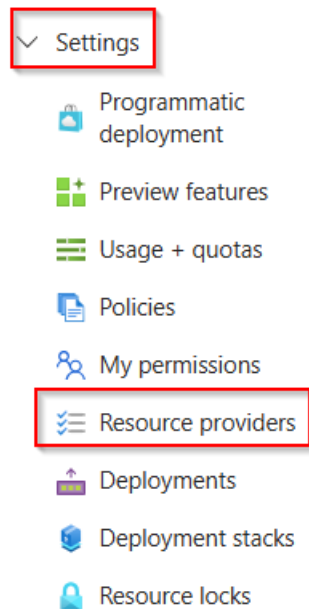
1. Sign in to the Azure portal and search for Subscriptions in the global search bar.



2. Select the subscription that requires the provider.

☐ Subscription name ↑↓	Subscription ID ↑↓	My role ↑↓	Current cost
☒ sub 12	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx	Owner	0.00
☐ sub 13	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx	Owner	0.00
☐ sub 14	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx	Owner	0.00
☐ sub 8	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx	Owner	0.00
☐ sub 9	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx	Owner	0.00

3. Under **Settings** in the left-hand navigation, select **Resource providers**.



4. Type the namespace into the filter box, for example **Microsoft.Quota**.

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quota	Status : All	Registration Policy : All
Provider ↑	Status	Registration Policy
○ Microsoft.Quota	... NotRegistered	RegistrationRequired ⓘ

5. Select the row for the provider, then choose Register from the command bar at the top of the blade.

Register

Unregister

Refresh

Export to CSV

Feedback

quota

Status : All

Registration Policy : All

Provider ↑	Status	Registration Policy
Microsoft.Quota	... NotRegistered	RegistrationRequired

The status column changes to Registering and then to Registered, normally within a minute or two. Refresh the page to confirm the final state before attempting a deployment.

3.4. Confirming the Registration State

A provider reports one of four registration states: NotRegistered, Registering, Registered, or Unregistering. Only Registered permits deployment of the namespace's resource types. Registration is asynchronous, so a state of Registering means the request was accepted and is still being applied; large namespaces such as Microsoft.Compute occasionally take several minutes. The Resource providers blade shows the current state for every namespace in the subscription, and the Status filter above the list can be set to Registered or Not registered to review the subscription's baseline.

3.5. The Microsoft.Quota Resource Provider and Subscription Quotas

Microsoft.Quota is the provider that backs the Quotas experience in the Azure portal and the Quota REST API. It exposes the current limit, the current usage, and the request interface for the quotas that Azure services enforce, including virtual machine cores by family and region, public IP addresses and other network limits, and machine learning compute. Until Microsoft.Quota is registered in the subscription, the Quotas page returns no data and calls to the Quota API fail with a MissingSubscriptionRegistration error, which means quotas can neither be viewed nor increased through the normal self-service path.

This matters because a newly created subscription frequently has Microsoft.Quota in the NotRegistered state. Nothing in a normal deployment forces the provider to register, so the gap is often discovered at the worst moment, when a deployment has already failed for lack of cores and the quota screen cannot be opened to fix it.

Once the provider reports Registered, the Quotas page lists the current limit and usage for each quota, filtered by provider, region, and subscription.

NOTE: Registering Microsoft.Quota does not change any limit by itself; it only makes the quota data and the request path available. Quotas are also scoped to a subscription and a region, so an increase granted in one region has no effect in another.

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IMPORTANT: A service whose own provider is not registered (for example Microsoft.Compute or Microsoft.MachineLearningServices) may not report its quotas until that provider is registered as well.

3.6. Troubleshooting

The table below lists the conditions most often seen during registration and the action that resolves them.

Symptom	Resolution
Deployment fails with MissingSubscriptionRegistration	Register the namespace named in the error message against the subscription that is the deployment target, then redeploy.
The Register command is unavailable or returns an authorization error	The account lacks the register permissions for the namespace. Request the Contributor or Owner role at subscription scope.
The provider remains in the Registering state	Registration is asynchronous. Wait several minutes and re-check the state; raise a support request if it does not settle.
The Quotas page is empty or reports an error	Microsoft.Quota is not registered in the selected subscription. Register it, then reload the page.

4. Definitions

Term	Definition
Resource provider	A service in Azure that supplies a set of resource types, identified by a namespace such as Microsoft.Compute or Microsoft.Quota.
Resource type	An individual object offered by a provider, written as namespace/type, for example Microsoft.Compute/virtualMachines.
Registration state	The status of a provider within a subscription: NotRegistered, Registering, Registered, or Unregistering.
Microsoft.Quota	The provider that exposes subscription quota limits and usage, and through which a quota increase is requested.

5. Associated Documents

- [Microsoft Learn: Azure resource providers and types](#)
- [Microsoft Learn: Quotas overview](#)

6. Document Approval

This document has been approved by:

- Manager, Cloud Sales Engineering

7. Document Revision History

Number	Effective Date	Revision History	Author
1.0	9/3/2026	Initial Draft	Matt Crumb

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8. Appendix A – Commonly Registered Provider Namespaces

The namespaces below are the ones most often required when a new subscription is prepared for a workload. The list is not exhaustive; the authoritative set for any subscription is the one shown on the Resource providers blade.

Namespace	Purpose
Microsoft.Compute	Virtual machines, scale sets, disks, and images.
Microsoft.Network	Virtual networks, load balancers, public IP addresses, and network security groups.
Microsoft.Storage	Storage accounts and their services.
Microsoft.Quota	Subscription quota limits, usage, and quota increase requests.
Microsoft.Insights	Azure Monitor metrics, alerts, and diagnostic settings.
Microsoft.ContainerService	Azure Kubernetes Service clusters.